

Policy Brief

Addressing the Skills Gap in Gilgit Baltistan: Aligning TVET with Labour Market Needs

Introduction

Gilgit-Baltistan is emerging as a region of strategic importance in Pakistan's development landscape. It is endowed with abundant natural resources, a pleasant climate, and a vibrant cultural heritage. It is home to diverse communities, scenic landscapes, and several districts where education levels are notably high, contributing to the region's growing potential in various sectors. The region holds significant potential for the benefits that can be realized through the effective implementation of TVET.

The shortage of a skilled workforce continues to pose challenges to inclusive and sustained economic growth. TVET-Cell in Gilgit-Baltistan has been making efforts to expand access to technical and IT education, however, issues such as limited rural outreach, inconsistent quality, and weak industry linkages persist. Reflections necessary to address these issues necessitate systematic study and stock-taking of the challenges and opportunities in the region for a sound policy foresight and effective interventions in future.

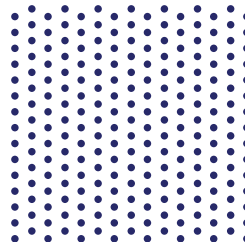
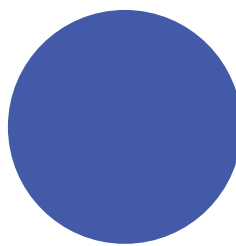
Conducted under the Team Europe-funded TVET Sector Support Programme (SSP) in collaboration with the National Vocational & Technical Training Commission (NAVTTTC), Pakistan, a report titled "Labour Market Assessment in Gilgit-Baltistan" identifies the critical gaps between workforce supply and demand across key sectors of the region. The study examined labour market trends, occupational demand supply gaps, and the relevance of existing TVET programmes. It engaged employers to identify skills mismatches and align training with industry needs. District-level mapping was undertaken to highlight labour shortages and surpluses, with a focus on high-demand occupations for disadvantaged groups, including women, persons with disabilities (PWDs), and transgender individuals. The findings aim to guide inclusive, market-responsive TVET planning and policy reform in Gilgit-Baltistan.

To have a sound research approach, the study employs both primary and secondary data collection methods, a robust and representative sample, and structured random sampling (SRS) and respondent-driven sampling (RDS) deriving a sample size of 1529. Proportional allocation across districts and targeted adjustments for small or strategically significant districts ensured equitable coverage and, therefore, reliable findings backing sound conclusions and recommendations.

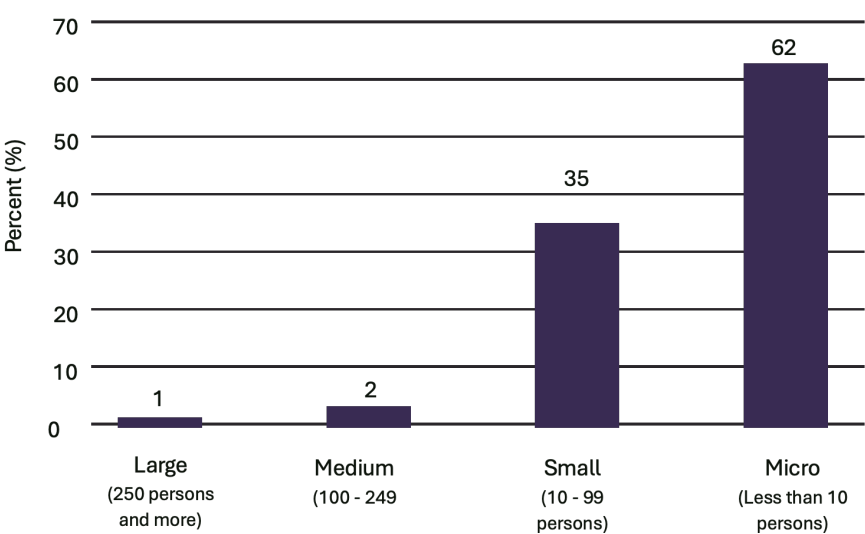
Survey Findings: Labour Market Assessment

The size-wise distribution of sampled establishments in Gilgit-Baltistan shows that micro-enterprises (fewer than 10 workers) dominate with 62% of the total, largely driven by informal, family-run, and agriculture-based activities. Small enterprises (10–99 workers) make up 35%, bringing the combined share of micro and small units to 97% of the industrial base. Medium-sized enterprises (100–249 workers) account for only 2%, while large firms (250+ workers) represent just 1%, a reflection of structural barriers such as remoteness, poor connectivity, infrastructure gaps, weak market access, and shortages of skilled labour.

Hospitality and Tourism represent the largest share of establishments at 19%, underlining the region's heavy reliance on its cultural heritage and natural tourist attractions. Allied Health (13%) and Agriculture, Forestry & Fishing



(12%) follow, illustrating the combined significance of healthcare provision and traditional livelihoods. Manufacturing (10%) and Construction (10%) contribute to industrial activity but remain limited in scale and hindered by skills shortages. Mining (8%), Automobile Repairing (6%), Printing & Packaging (6%), and Handicrafts (5%) continue to provide important local employment. However, growth is constrained by weak market linkages and limited capacity to scale. Emerging areas such as Water (5%) and Renewable Energy (3%) signal gradual diversification, while Other Services, at just 3%, indicate their relatively minor role in the regional economy.

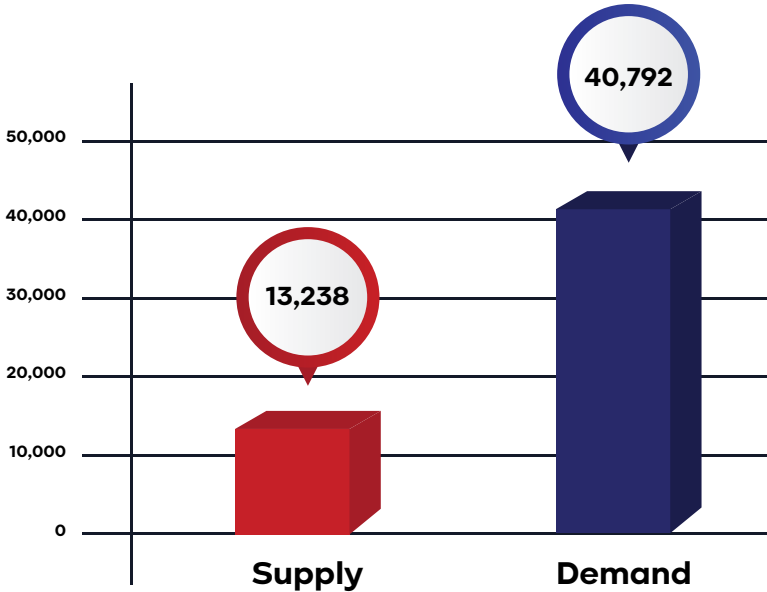


Annual Skilled Workforce Demand and Supply

The labour market in Gilgit-Baltistan shows a stark gap between workforce needs and needs met. The demand for skilled workers is around 40,792, while the current supply reaches only 13,238. This leaves 27,554 positions unfilled – about 68% of the total. Such a gap limits the region’s ability to meet industry needs, increases reliance on workers from outside, and pushes many into informal jobs. The 3:1 ratio of demand to supply signals the urgent need to strengthen and modernize technical and vocational education & training (TVET) systems.

Looking at specific trades, there are strong mismatches between the types of skills in demand and those being produced. Jobs such as Cooks, Software Developers, and Nurses are in very high demand, especially in Tourism, ICT, and Healthcare. On the other hand, many young people are still being trained in traditional occupations that no longer match market realities, creating an oversupply in less relevant fields. This shows the need to update training programmes and link them better with actual employment opportunities.

Sectors with the greatest growth potential – Tourism, Renewable Energy, and Agriculture – require specialised skills that are not yet being addressed effectively through current training initiatives. Building stronger partnerships with employers, promoting apprenticeships and internships, and expanding on-the-job-trainings will be essential to prepare workers for these opportunities. Without timely action, Gilgit-Baltistan may struggle to realise its economic potential by providing sustainable livelihoods for its people.



Sector-Wise Skilled Workforce Demand

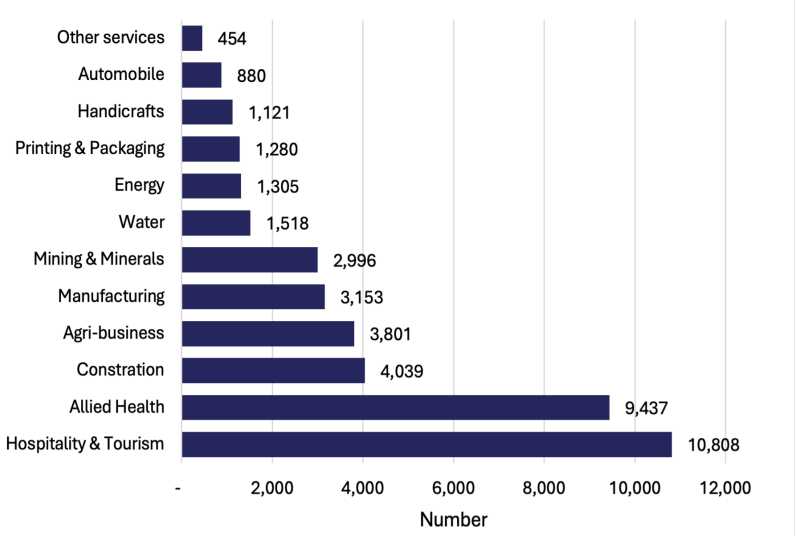
With 10,808 people needed, the hospitality and tourism sector in Gilgit-Baltistan has the highest sector-wise demand for skilled labour, indicating the region’s strong reliance on its tourism potential and cultural assets. Allied Health has the second highest demand with 9,437 employees needed, highlighting the province’s urgent need for support and medical personnel. Agribusiness (3,801) and construction (4,039) exhibit significant demand as well, propelled by the progressive industrialization of agriculture and continuous advancement needs of the infrastructure.

Demand is relatively moderate but significant in other sectors. While Manufacturing, Mining, and Water Services continue to offer job opportunities, the need for 1,305 personnel in Renewable Energy sector indicates an increasing trend toward sustainable energy sources. Despite their modest scale, traditional industries like Printing and Packaging (1,280) and Handicrafts (1,121) continue to be important for local livelihoods. Furthermore, Gilgit-Baltistan’s labour market is diverse as seen by the various services (454) and automobile repair (880) sectors.

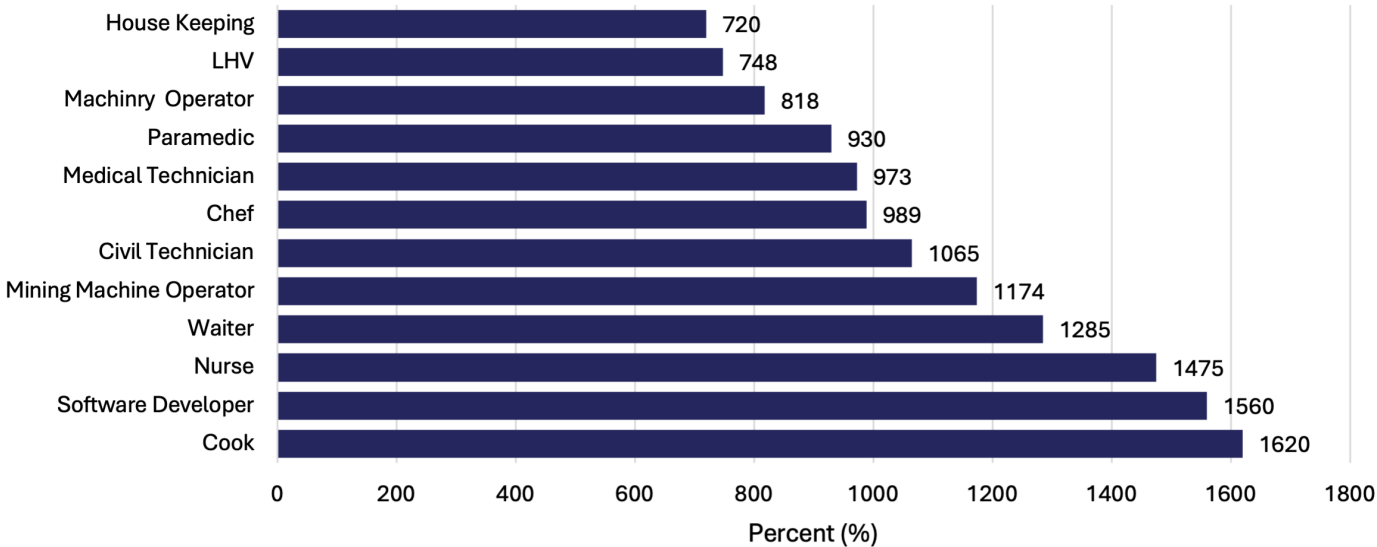
When taken as a whole, these numbers highlight the necessity of a multi-sectoral TVET strategy that strikes a balance between skill development in established and emerging professions, as well as investment in key industries, like Tourism and Healthcare. This strategy will help boost inclusive growth, improve worker preparedness, and fortify the area’s economic resilience.

Trade-Wise Skilled Workforce Demand

The survey highlights that Gilgit-Baltistan’s Hospitality and Tourism sector employs 10,808 workers against a strong demand for skilled labour in the region’s tourism industry. Demand of Healthcare roles remains high, with 9,437 Allied Health positions, while demand for roles in Agribusiness (3,801) and Construction (4,039) seem to be undergoing a steady growth. Among individual trades, Cooks are most sought after (1,620 posts), followed closely by demand of 1,560 Software Engineers, reflecting both food service needs and the expansion of ICT.



Within Tourism, food services are central exhibiting a demand for 989 chefs and cooks. At the same time, ICT jobs such as software developers show the region’s advancing digital economy (1,560) alongside traditional industries. Healthcare personnel shortages are evident with demands for 748 female health visitors, 973 medical technicians, 930 paramedics, and 1,475 nurses, underscoring the urgency to improve access to health services in remote communities and to strengthen human resource for equitable healthcare.



District-Level Skills Landscape

Gilgit-Baltistan’s district-level skills landscape demonstrates the region’s varied needs for skilled workers in Mining, Construction, Healthcare, and Hospitality. Training institutions like the Government Polytechnic Institute and AKRSP centres provide courses in electrical work, hospitality, IT, and renewable energy. The high-demand jobs include paramedics, civil technicians, welders, and mining machine operators.

Skardu’s mining, tourism, agricultural, and healthcare industries all have a great impression on GB’s labour market. Cooks are in the greatest demand, underscoring the significance of food services and hospitality in the region. By offering relevant courses, the Skardu Technical Training Institute and vocational centres give access to trainings on Both internal and foreign travel have a significant impact on Hunza’s workforce demand, opening doors for jobs in the hospitality and culinary industries. Aga Khan centres and private institutions that offer courses in tourism, IT, small business, and handicrafts provide necessary technical assistance in major districts.

With the help of a few vocational centres that offer instruction in operating heavy machinery, Diamer’s demand is

mostly driven by the Mining and Construction industries. Ghizer offers more trainings in forestry, beekeeping, and agriculture, and there is a high demand for skills education in the healthcare and agricultural industries. While districts like Nagar, Ghanche, Kharmang, and Shigar rely mostly on unofficial and community-based skill development.

Astore has shown promise of growth in the areas of renewable energy and livestock management that needs to be reflected in the skills and training of work force in the district.

Skills, Employment, and Marginalised Groups

Analysis by gender trends in GB's labour market reveals notable differences in provisions for male and female participation, with women only having access to 23.4% of all skilled employment. Construction, Energy, and Mining are still heavily dominated by men, although areas like Allied Health (4,398 out of 9,437) and Hospitality (1,736 out of 10,808) are relatively more gender inclusive. The sociocultural dynamics of the province, which still limit the enabling conditions for women in technical occupations, are reflected in this discrepancy. However, some districts of the region, such as Skardu, have completely different trends of gender parity reflecting in higher gender parity in education and workforce.

Figures on underprivileged populations in Gilgit-Baltistan show enhanced, although not ubiquitous, efforts towards making the region's economy more inclusive. Minorities are beneficiaries to 39% of the available opportunities, followed by people with disabilities (36%) and older people (18%). Nonetheless, the majority of these groups continue to be restricted to low- to semi-skilled jobs, such as machine operation, basic administrative assistance, and janitorial work, indicating limited access to specialist occupations or career paths with higher upward mobility.

Recommendations

The Labour Market insights for Gilgit-Baltistan point to the need for urgent reforms in curricula, industry engagement, and the geographical expansion of training programmes aligned with market realities, while ensuring inclusivity and sustainability.

Some specific recommendations are as below:

- Establish fully equipped multipurpose TVET centres in each district of GB, aligned with local economic strengths (e.g., tourism, agribusiness, energy, construction), to widen access beyond Gilgit and Skardu.
- Set up women-only training wings with female instructors, flexible hours, and safe facilities to boost female participation in high-demand, non-traditional trades while respecting cultural norms.
- Reform TVET financing by introducing output-based models that reward institutions for placements, female/PWD enrolment, programme completion, and international mobility outcomes.
- Revise and modularize curricula in collaboration with industry to strengthen practical competencies in emerging sectors such as renewable energy, hospitality, digital freelancing, and construction technology.
- Improve governance structures at the district level by establishing TVET–Industry Coordination Committees with employers, training providers, and community representatives to ensure rapid responsiveness to local workforce needs.
- Strengthen oversight bodies (TEVTA, NAVTTC GB regional office) through digital MIS, placement tracking, performance-based budgeting, and transparent monitoring and evaluation (M&E) systems.
- Develop specialized training tailored to GB's terrain and risks (e.g., earthquake-resistant construction, flood-resilient housing, smart infrastructure, CPEC-related civil technologies).
- Align selected TVET programmes with international certifications (e.g., City & Guilds, ISO) to facilitate labour mobility to GCC countries and participation in mega-projects.
- Launch mobile training units to deliver short, trade-specific certification programmes in underserved and remote valleys, targeting youth and disadvantaged populations.
- Introduce Recognition of Prior Learning (RPL) frameworks to certify informal workers and support their transition into formal employment.
- Promote public-private partnerships (PPPs) to establish sector-specific Centres of Excellence, such as a Hospitality Skills Hub in Skardu, a Renewable Energy Lab in Hunza, and a Gemstone Centre in Nagar, to drive advanced training.